Asmita Pal

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Summary

Performance Architect at AMD's Data Center GPU team, driving performance modeling for next-generation GPUs targeting cloud, ML and HPC workloads. Prior work in PhD involved privacy-preserving traces, fully homomorphic encryption, and interconnect security.

Education

PhD in Computer Engineering — August 2018 – August 2025
University of Wisconsin - Madison

MS in Computer Engineering — August 2016 – August 2018
Utah State University

B.Tech. in Electronics and Instrumentation Engineering — July 2010 – June 2014
West Bengal University of Technology, India

Publications

- J Kalamatianos, JB Kotra, **A Pal**: **Managing a Cache Using Per Memory Region Reuse Distance Estimation**, US Patent App 20240211407
- **Asmita Pal**, Keerthana Desai, Rahul Chatterjee, Joshua San Miguel: **Camouflage: Utility-Aware Obfuscation for Accurate Simulation of Sensitive Program Traces**, published in ACM Trans. Archit. Code Optim. 21, 2, Article 36 (June 2024), 23 pages.
- Mitali Soni, **Asmita Pal**, Joshua San Miguel: **As-Is Approximate Computing**, published in ACM Trans. Archit. Code Optim. 20, 1, Article 3 (March 2023), 26 pages.
- **Asmita Pal**, Karthik Swaminathan, Subhankar Pal, Joshua San Miguel: **Characterizing Memory side channels in FHE applications** presented in DISCC Workshop *(held in conjunction with MICRO'22)*
- Pramesh Pandey, **Asmita Pal**, Koushik Chakraborty, Sanghamitra Roy: **Reliability and Uniformity Enhancement in 8T-SRAM based PUFs operating at NTC**, published in IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED) 2018.
- **Asmita Pal**, Aatreyi Bal, Koushik Chakraborty, Sanghamitra Roy: **Split Latency Allocator: Process Variation-Aware Register Access Latency Boost in a Near-Threshold Graphics Processing Unit**, published in J. Low Power Electronics 13(3): 419-427 (2017).

Work Experience

AMD — Santa Clara, United States
Memory Systems Performance Architect — September 2025 – present

- Working with the Data Center GPU team to optimize high-performance SoCs for Cloud computing, and machine learning acceleration. My role focuses on performance modeling, and analysis for interconnects in next-generation GPU systems.

AMD — Boston, United States
Research Intern — June 2022 – Sept 2022

- Characterized last-level cache (LLC) traffic for AI and HPC workloads on multicore CPU SoCs to uncover access patterns as performance bottlenecks.
- Explored reuse-distance guided heuristics and ML-based approaches to inform adaptive cache replacement strategies.
- Contributed to a US patent on dynamic cache management using per-region reuse distance estimation.

AMD — Boston, United States

Research Intern — September 2020 – December 2020

- Profiled memory and LLC behavior using hardware performance counters to analyze BERT inference kernels.
- Evaluated software-hardware interaction metrics to identify performance bottlenecks and acceleration opportunities in memory-bound components.
- Generated kernel-specific traces to guide system-level optimizations targeting memory intensity and execution hotspots.

Infosys Technologies Limited — Pune, India

System Engineer — July 2014 – August 2016

- Developed an automation framework for hardware verification, streamlining debugging processes and reducing manual verification time, to enhance team productivity.
- Managed client deadlines and service delivery, coordinating with cross-functional teams.

Research Experience

STACS Lab, University of Wisconsin - Madison — Madison, WI

Graduate Research Assistant — August 2018 – August 2025

Advisor: Joshua San Miguel

Side-channel attacks for mesh-based Network-on-Chips (NOC)

- Modeled synthetic traffic to observe contention behavior in mesh-based NoCs using Garnet in gem5.
- Analyzed performance bottlenecks through traffic pattern analysis for existing temporal domain-partitioning schemes, such as SurfNoC.
- Exploring novel time-sharing schemes to mitigate side-channel vulnerabilities arising due to contention, without compromising security.

Fully Homomorphic Encryption (FHE)

- Developed analytical models for performance prediction of kernels present in FHE-based applications using OpenFHE library based on CKKS schemes. Characterized workloads such as encrypted matrix multiplication, logistic regression, ResNet20, CIFAR10 and Chi-Square GWAS protocols.
- Analyzed side-channels in applications encrypted using FHE (Fully Homomorphic Encryption), on the IBM HELib framework.
- Modeled spatial/temporal access patterns across >300K encrypted credit card transactions (Kaggle dataset), achieving 90% classification accuracy in side-channel leakage using ML classifiers.
- Distinguished spatio-temporal vectors of unique entries using machine learning classifiers with up to ∼90% accuracy revealing a critical vulnerability in what is considered the gold standard of security.

Camouflage: Privacy-preserving Traces

- Addressed challenges in sharing confidential workloads for architectural research by balancing trace utility and privacy.
- Proposed an obfuscation mechanism for memory traces from SPEC2017, GAPBS and PERFECT Benchmarks.
- Characterized utility of a trace as a function of program semantics that influence program behavior, as captured in memory traces.
- To ensure input privacy, developed a threat model based on input indistinguishability and showed that the average security loss is 7.8% for obfuscated traces.
- Tested camouflaged encryption (AES, DES, Blowfish) traces against secret key recovery.
- Validated these traces against prefetchers and replacement policies on ChampSim exhibiting an average τ correlation of 0.6, given $\tau \in [-1, 1]$.

Prefetching using Large Language Models

- Analyzed spatial and temporal patterns to generate spatio-temporal vectors for studying address patterns.
- Applied transformer-based models to memory access sequences for predictive address prefetching.
- Compared heuristic vs LLM-predicted addresses for coverage and accuracy.

Approximate Computing

- Applied reduced precision, random input sampling and tree-based sampling techniques for approximating applications used in image processing/computer vision.
- Developed error metrics for assessing completeness of graph-based algorithms undergoing approximation.
- Simulated these approximate PERFECT and GAPBS Benchmarks on a speculation-based architectural simulator for studying accuracy-runtime trade-offs.
- *Part of this work was recognized as winner of the N+1 Institute x Google Reverse Pitch Competition on AI Sustainability at UW-Madison.*

Instructor: Parmesh Ramanathan

Digit Recognition on FPGA using Multilayer Perceptron

- Designed processor, memory and accelerator modules to enable storage of a pre-trained model and offload computations to accelerator.
- Captured image and predicted image digit on the 7-segment display.

Instructor: Younghyun Kim

Precise encoding in Stochastic Computing for Neural Networks

- Proposed a new encoding scheme for Stochastic Numbers and implemented this new encoding format in MAC module of a small-scale Neural Network.
- Analyzed the energy-accuracy trade-off for LeNet300-100 and observed up to 5X energy improvement.

Bridge Lab, Utah State University — UTAH, UNITED STATES

Graduate Research Assistant — August 2016 – July 2018

Advisors: Koushik Chakraborty and Sanghamitra Roy

Design of a Process-Variation Aware Low Power GPU

- Explored the role of process variation on register files for a GPU operating at Near Threshold Voltages.
- Analyzed the performance variation in GPGPU applications from AMD APP SDK suite, with varying register latency.
- Proposed a low-overhead technique to overcome the effects of process variation on register latency. Exhibited ~36% reduction in GPU energy consumption with marginal area and power overheads.

Speaker Diarization using Deep Neural Networks

- Extracted features from a conversation and used Deep Neural Networks to separate an audio signal according to speaker identity.
- Compared accuracy of the speaker diarization framework, using different deep neural networks (DNN) and convolutional neural networks (CNN).

Critical Thread Predictor

- Studied the statistical correlation of thread criticality metrics based on cache-misses and synchronization bottlenecks.
- Analyzed performance numbers for Splash2 Benchmarks on multi-core Sniper simulator.

Awards

- Winner of N+1 Institute X Google Reverse Pitch Competition - (AI Sustainability).
- Sarah and David Epstein Teaching Fellowship
- Student Research Grant [Awarded by Office of Diversity, Inclusion and Funding at **UW-Madison**]
- ISCA 2022 Travel Grant
- Anita Borg Grace Hopper Celebration 2019 Scholarship
- Wisconsin Distinguished Fellow - Schneider
- Rajendra Prasad and Yasoda Mani Grandhi Fellow

Research Interests & Technical Skills

Research Interests : Computer Architecture, System Performance Modeling, Memory Subsystems, Hardware-Software Co-design, Secure and Private Computing, Approximate Computing, Workload Characterization, Trace-based Simulation, NoC Security

Programming Languages : C++, Python, Bash, Verilog

Architectural Tools : ChampSim, Gem5, Garnet, PinTool, Perf, likwid

Other Libraries : Tensorflow, HELayers, OpenFHE

Relevant Coursework

Advanced Computer Architecture I and II, Introduction to Operating Systems, Neural Networks, Topics in Security and Privacy, Introduction to Programming Languages and Compilers, Embedded Computing Systems

Professional Service

Invited Talks

- Town Hall - University of Wisconsin-Madison, Dept. of Electrical and Computer Engineering, February 2025
- Computer Architecture Coordination Seminar Series - University of Central Florida, November 2023

Artifact Evaluation Committee

- IEEE/ACM International Symposium on Microarchitecture 2023 (MICRO 2023, 2022)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2022)
- Conference on Machine Learning and Systems (MLSys 2020)

- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2020)

Teaching

ECE 757 (Advanced Computer Architecture II)	Spring 2024, Spring 2025
ECE 352 (Digital System Fundamentals)	Fall 2024
ECE 5750 (High Performance Computer Architecture)	Fall 2017